



Evolutionary insight and characterization of *WOX* genes in callus development and differentiation of *Peucedanum praeruptorum*

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Abstract

Main conclusion The *WOX* genes essential for callus formation and development were successfully isolated from *Peucedanum praeruptorum*, confirming the involvement of *PpWOX9*, *PpWOX14*, and *PpWOX15* in this process.

Abstract *WUSCHEL* (*WUS*)-related homeobox (*WOX*) gene family is essential due to its pivotal role in orchestrating fundamental developmental processes, such as stem cell maintenance, organ morphogenesis, and embryonic patterning, which underpin plant growth and adaptability. They form intricate regulatory networks to ensure cellular homeostasis and enable indeterminate growth, revealing the evolutionary innovations and how plants achieve complex architectures. However, the specific role of *WOX* members in regulating callus development and differentiation in *P. praeruptorum* remains unclear. Using bioinformatics, genomics, and transcriptomics methods, we performed a systematic analysis of the structural composition, family classification, evolutionary relationships, and expression patterns of *WOX* genes in *P. praeruptorum* during callus formation. *PpWOX* proteins were categorized into three main subclades based on the phylogenetic tree. *Cis*-acting element analysis indicated that many *PpWOX* genes possess multiple copies of light-responsive elements (LREs), suggesting that light signals significantly influence the expression of these genes. Synteny analysis revealed eight whole-genome duplication (WGD) and segmental duplication events in their evolutionary history, which may have facilitated the expansion of *PpWOX* genes. The expression profile showed distinct expression levels for *PpWOX* genes during callus formation, with *PpWOX9*, *PpWOX13*, *PpWOX14*, *PpWOX15*, and *PpWOX16* exhibiting higher expression levels compared to other *PpWOX* genes. qRT-PCR analysis demonstrated that the expression levels of *PpWOXs* varied across six different stages. During the somatic cell development stage, the expression levels of *PpWOX9* and *PpWOX15* were significantly elevated. Transcriptional co-expression analysis revealed that *PpWOX9*, *PpWOX14*, and *PpWOX15* exhibited similar co-expression patterns. Subcellular localization studies using tobacco protoplasts showed that the *PpWOX9* and *PpWOX15* genes were specifically localized in the nucleus. Our findings serve as a scientific reference for the study of *WOX* genes in the conservation of *P. praeruptorum* resources.

Keywords *WUSCHEL*-related homeobox · Callus formation · Somatic embryo morphogenesis · *Peucedanum praeruptorum*

Introduction

Peucedanum praeruptorum Dunn is a perennial, once-flowering plant in the Apiaceae family. Coumarin compounds are considered the main active ingredients in

Peucedanum species (Song et al. 2022, 2023). Currently, more than 80% of the domestic production of *P. praeruptorum* is produced by artificial cultivation, while it still generally relies on seed propagation (Song et al. 2021; Pan et al. 2024). Due to lignification and a decrease in the concentration of medicinal components, the roots lose their potential as materials for traditional Chinese medicine after bolting (Song et al. 2024). The artificially cultivated *P. praeruptorum* is often harvested in the autumn and winter of the same year or before bolting in the second

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year. The insufficient growth years lead to an insufficient accumulation of medicinal coumarins in the roots. Currently, material quality and germplasm degradation are becoming increasingly prominent, and seed propagation cannot completely solve the early bolting issue. Cultivating late-bolting varieties or adopting chemical measures to delay bolting will facilitate improving this situation.

The WOX family is a plant-specific transcription factor that belongs to a large family of homeobox (HB) transcription factors. HB proteins were first identified in *Drosophila*, and their typical homeobox domain (HD) consists of 60–66 amino acid residues in a special spatial structure, which forms a “helix-loop-helix-turn-helix” configuration by folding (van der Graaff et al. 2009; Romani et al. 2018). In higher plants, WOX family members play an important role in callus formation, embryonic development, stem cell maintenance, differentiation of lateral organs such as rhizomes, seed germination, and tissue and organ regeneration (Hao et al. 2019). Since the first identification of WOX genes in *Arabidopsis thaliana* in 2004, WOX family members in many plants have been identified recently (Haecker et al. 2004). The WOX family of the dicotyledonous plant *A. thaliana* contains 15 members, which are divided into three branches according to the phylogeny: the modern (WUS) clade, the intermediate clade, and the ancient clade (Lin et al. 2013). In the modern branch, AtWUS and AtWOX 1–7 are clustered. AtWOX 8, 9, 11, and 12 are grouped together on the intermediate branch, and AtWOX 10, 13, and 14 are grouped together on the ancient branch. The homeodomain-conserved domain of the WOX protein contains 65 amino acid residues.

Recent forward and reverse genetic studies have shown that plant WOX members play a role in various physiological and embryonic developmental processes. Several AtWOX genes are essential for embryonic patterning, stem cell maintenance, and organ formation (Ma et al. 2019). AtWUS maintains stem cell homeostasis in the shoot apical meristem (SAM) during all the developmental stages, while in WUS-knockout mutants, SAM maintenance is disrupted both in the embryonic and post-embryonic stages (Satterlee et al. 2023). AtWOX2 is expressed in the egg cell and zygote, but it is restricted to the apical daughter cell of the zygote through asymmetric division, which is required for the normal development of the zygote apical cell (Zhao et al. 2015a, b). PRESSED FLOWER1 (PRS1/AtWOX3), a different member of the AtWOX family, is found around the shoot meristem. PRS1/AtWOX3 recruits founder cells from all meristem layers to form the lateral domains of vegetative organs and floral organs (Sarkar et al. 2007). AtWOX5, a homolog of AtWUS, is expressed in the quiescent central cell of the root and functions to maintain stem cells in the root apical meristem (Park et al. 2005). A few SEEDS2 (PFS2/

AtWOX6) genes regulate ovule development, and their transcripts accumulate primarily in developing ovules. In pfs2-mutants, ovules and embryo sacs develop abnormally, and pfs2 loss leads to decreased fertility (Wu et al. 2005). During vegetative development, STIMPY (STIP/AtWOX9) plays an important role in SAM growth. It maintains cell division and prevents premature differentiation in certain SAM and root system tissues (Schlegel et al. 2021). Nardmann and Werr cloned two WUS homologs in maize (ZmWUS1 and ZmWUS2) and one WUS homolog in rice (OsWUS) and found that they were not expressed in the organizing center of the vegetative SAM as in *Arabidopsis* (Chu et al. 2013). The specific homeobox gene QHB/OsWOX5 is a homolog of AtWOX5 in rice, expressed in the quiescent center, and has similar stem cell maintenance abilities (Zhao et al. 2009). OsWOX11 is a rice homolog of AtWOX11 that can activate crown root germination and growth (Atta et al. 2009).

The WOX family genes exhibit spatiotemporal expression specificity during various stages of callus induction and differentiation. The pluripotent callus is formed by dividing xylem pole pericycle cells. The process is similar to that of lateral root formation. Those genes that are involved in lateral root formation are also implicated in the formation of pluripotent callus (Kim et al. 2018). The expression of some genes that mark the root meristem, including WOX5, SHR, and root clavata homolog 1 (RCH1), was significantly up-regulated. The transcription of WOX5, SCR, PLT1, and PLT2 is activated by HAT proteins from the GNAT/MYST superfamily, which directly bind to their promoters to initiate acetylation (Kareem et al. 2015). In tissue culture, nascent root organogenesis is initiated by transferring pluripotent callus to an auxin-rich root induction medium. Auxin rapidly induces the expression of PLT3, PLT5, and PLT7, while also indirectly regulating the transcription of PLT1 and PLT2 (Su et al. 2020). WOX11 promotes pluripotency acquisition by activating lateral organ boundary domain 16 (LBD16), which in turn promotes the expression of WOX5, PLT1, and PLT2 through the activation of ARFs (Xu and Huang 2014).

In recent years, pluripotent callus transcriptome, epigenome, and cell line analyses have shown that the formation of pluripotent callus and the development of new root organogenesis share similar genetic pathways (Druege et al. 2016). De novo root organogenesis can be divided into two steps: the first involves the transformation of energetic cells into root establishment cells, which is marked by the WOX11 gene; the second involves the transformation of root establishment cells into root primordium cells, marked by the WOX5 gene. Inhibition of polar auxin transport blocks the rooting process, indicating that auxin is a key hormone regulating the development of new root organogenesis (Pan et al. 2019). By inhibiting the bioproduction of auxin mediated by YUC, it suppresses the expression of WOX11 and ultimately prevents the fate transition of susceptible cells (Long et al. 2022). Auxin first directly activates the expression of

WOX11 and its homolog, *WOX12*. Subsequently, *WOX11* and *WOX12* promote the expression of *WOX5* and *LBD16*. Additionally, auxin induces the expression of *PLT3*, *PLT5*, and *PLT7*, thereby regulating the expression of downstream root meristem marker genes (Hu and Xu 2016).

At present, the production and seedling propagation of *P. praeruptorum* urgently require a new propagation method. Establishing an in vitro regeneration system for *P. praeruptorum* through tissue culture is an effective approach for cultivating late-bolting, high-quality resources, as well as enhancing the conservation and improvement of germplasm resources (Pan et al. 2024). As a type of tissue-specifically expressed gene, *WOX* plays a key role in regulating meristem development and cell differentiation. Based on our established tissue culture system, genome-wide screening and functional identification of *WOX* genes will help us transform specific *WOX* genes in *P. praeruptorum* to exert their effects. Here, bioinformatics, genomics, and molecular biology methods were integrated to systematically analyze *PpWOX* formation, gene duplication, systematic evolution, and tissue-specific expression patterns, providing a new strategy for the conserving and improving *PpWOX* genes in germplasm resources.

Materials and methods

Plant materials and sample collection

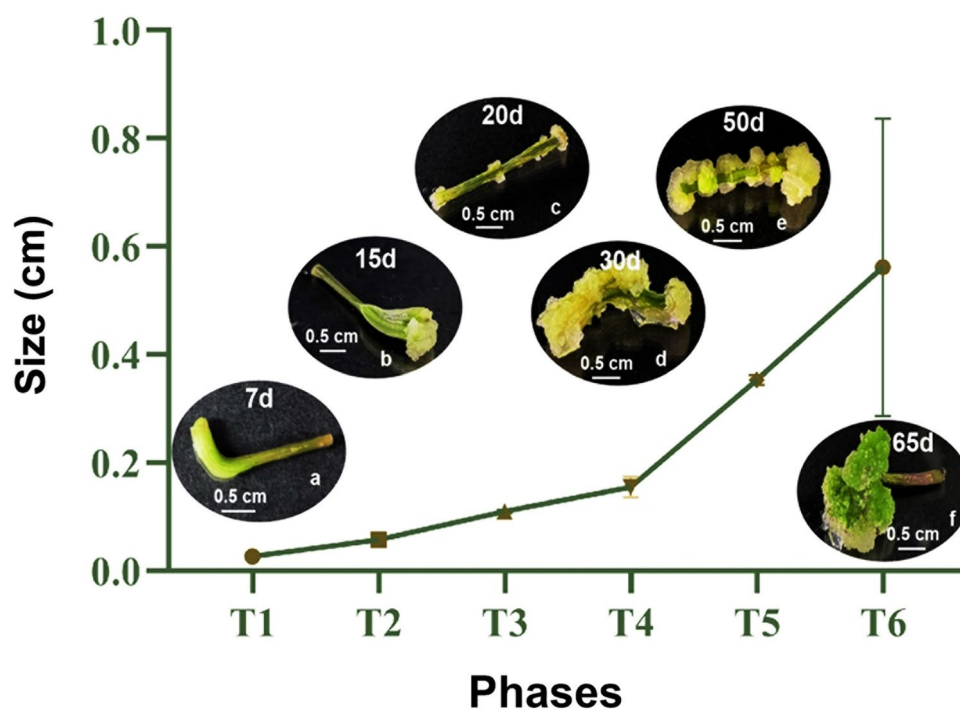
The *P. praeruptorum* seedlings used for explants were from the medicinal plant garden of West Anhui University (Luan,

China). The optimized MS medium with 0.5 mg/L BA and 2 mg/L 2, 4-D was used for the induction of *P. praeruptorum* callus. The explants were selected from the stem sections of *P. praeruptorum*. During the induction culture for approximately 65 days, the callus formation were divided into 6 stages (Fig. 1). The T1 stage refers to the stem segment explant that has been cultured in the dark for 7 days, during which no callus is induced. The T2 stage refers to the explant that has been cultured in the dark for 7 days and in the light for 8 days, during which the explants swell and form calluses at both ends. The T3 stage describes the callus that has been cultured for 20 days, by which time all the wounds of the explants have induced and the callus at both ends begins to develop. The T4 stage occurs when the callus has been cultured for 30 days. The T5 stage is reached when the callus has been cultured for 50 days, at which point it begins to transform into somatic embryos. The T6 stage refers to the tissue that has differentiated into globular embryos. The callus samples taken at different stages were frozen in liquid nitrogen and stored at -80°C . A total of 0.5 g of callus was collected at each stage, and three biological replicates were performed.

Identification and physicochemical property analysis of the *WOX* members of *P. praeruptorum*

The genome files, annotation files, protein sequences, and CDS files of *P. praeruptorum* were downloaded from Figshare (<https://doi.org/10.6084/m9.figshare.21743984.v1>). The HMMER 3.0 program was used to preliminarily screen all of

Fig. 1 Callus tissue morphology of *P. praeruptorum* at different stages. T1 to T6 represent the 7th, 15th, 20th, 30th, 50th and 65th day from explant inoculation to callus induction, respectively



the WOX protein sequences based on the hidden Markov model of the WOX conserved domain (PF00046) and seed files from InterPro (<https://www.ebi.ac.uk/interpro/entry/pfam/PF00046/>). The conserved domains of the selected WOX proteins were verified using the CDD database (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) and the InterPro database (<https://www.ebi.ac.uk/interpro/>). Finally, sixteen WOX family members were identified from the *P. praeruptorum* genome. The ProtParam tool (www.web.expasy.org/ProtParam) was utilized to analyze the physicochemical properties of the protein sequences belonging to the WOX gene family members. The Cell-PLoc-2 (www.csbio.sjtu.edu.cn/bioinf/Cell-PLoc-2) was employed to predict the intracellular localization of the *PpWOX* genes within *P. praeruptorum* cells.

Multiple sequence alignment and phylogenetic tree

Protein sequences from Arabidopsis, corn, rice, and tobacco that are part of the WOX family were downloaded from PlantTFDB v5.0 (www.planttfdb.gao-lab.org). The protein sequences of the identified WOX members were aligned using ClustalW software. The phylogenetic tree was constructed using the neighbor-joining (NJ) method in MEGA X software, with the bootstrap set to 1000 and other parameters set to default. The constructed phylogenetic tree was beautified using the iTOL web service (www.itol.embl.de).

Gene structure, conserved motifs, and conserved domain analysis

The motif analysis of WOX gene family members was performed using the MEME suite tool (www.meme-suite.org/meme/tools/meme). The number of motifs was set to 15, and the motif length ranged from 6 to 200. The CDD database at NCBI was used to obtain the conserved structures of the WOX family members. The gene structure was visualized using TBtools software.

Cis-acting element analysis

TBtools software was used to obtain the promoter sequence located 2000 bp upstream of the *PpWOX* genes. Simultaneously, the online resource PlantCARE (www.bioinformatics.psb.u-gent.be/web-tools/plantcare/html/) was used to predict the *cis*-acting elements. The number of promoter elements present in the 16 WOX members was quantified, and heat maps were generated and classified.

Chromosome mapping, synteny, and selection pressure analysis

To visualize the chromosome location of the identified *PpWOX* family members, TBtools software was used. The

Multiple Collinearity Scan Toolkit (MCScanX) was used to employed to analyze the collinearity of the *P. praeruptorum* genome and all gene duplication events. *A. thaliana*, *P. praeruptorum*, and *Angelica sinensis* were selected to examine the collinearity among these three species. TBtools software was also used to screen the genome length, annotation information, and tandem repeat gene pairs, as well as to visualize the tandem repeat and chromosome locations of WOX members. The Ka/Ks ratio was served to evaluate the selection pressure on *PpWOX* genes. To determine the divergence time of *PpWOX* genes, the following formula was used: $T = Ks/2$, where $\lambda = 6.56 \times 10^{-9}$ for monocot.

Data quality control and quantitative analysis of transcriptome sequencing

The total RNA from the *P. praeruptorum* tissues at different stages was extracted by the TRIzol extraction method. The purity, concentration, and integrity of the RNA were detected by the NanoDrop 2000 spectrophotometer and the Agilent 2100/LabChip GX. After the sample was qualified, the cDNA library was constructed through mRNA enrichment and fragmentation, mRNA reverse transcription, adapter addition, and PCR enrichment. After the library was constructed, the Qubit 3.0 fluorescence quantification instrument was used for preliminary quantification, and the concentration was required to reach more than 1 ng/ul. Then, the Qsep400 high-throughput analysis system was used to detect the inserted fragments of the library. When the inserted fragments met expectations, the library's effective concentration (> 2 nM) was accurately quantified using the Q-PCR method. After the library quality inspection was qualified, the Illumina HiSeq 2500 platform was used for sequencing. Trimmomatic was utilized to filter the raw sequencing data (raw reads) by removing sequences that contain adapters, sequences including poly-N, and low-quality sequences in order to obtain valid data (clean reads). Subsequently, FastQC was employed to calculate the Q20, Q30, GC content, and sequence duplication level of the clean reads. Hisat 2 and StringTie software were used to align and assemble the validated data with the reference genome of *P. praeruptorum*. For quantitative expression analysis, FPKM (fragments per kilobase of exon model per million mapped fragments) was employed for normalization, serving as an indicator to assess the expression levels of transcripts or genes. The FPKM values for gene expression were obtained through transcriptome sequencing (accession number: PRJNA1083043). Following $\log_2$ (FPKM + 1) normalization, TBtools software was utilized to visualize the expression profile.

Screening and functional enrichment of differential expressed genes

The DESeq 2 program was used to perform gene differential expression analysis on samples from the T2 and T6 stages. The false discovery rate (FDR) was determined by applying multiple hypothesis test corrections based on the significance of the differences. A fold change ≥ 1.5 and P -value < 0.01 were established as the screening criterion for identifying differentially expressed genes. EggNOG-mapper was employed to perform Gene Ontology (GO) annotation and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis on the differentially expressed genes obtained, to further evaluate the functions of these genes (Table 1).

Weighted gene co-expression network analysis

The goodSamplesGenes function in the WGCNA package was used to detect and filter genes with missing values, while the remaining genes were employed for WGCNA analysis (Langfelder and Horvath 2008). After importing the *R* package (v1.70–3) in *R* Studio (v4.0), the constructed expression matrix and trait matrix were loaded into the program. The samples in the expression matrix were clustered based on their expression levels. The weight value was determined using the soft threshold method. A scale-free network was generated using the one-step “block-wise Modules function,” with power chosen from the previous step, the minimum number of genes in the module set to 40, the merge module parameter adjusted to 0.25, the maximum number of genes in the module capped at 20,000, and all the other parameters set to their default values. The co-expression module and the trait matrix were analyzed using a labeled heatmap. We screened out the functional genes that coexpressed with the *WOX* gene by using the export network function in Cytoscape. The co-expression network was visualized using Cytoscape software, and the co-expressed genes were imported into *R* Studio for GO and KEGG annotation visualization.

Real-time fluorescence quantitative PCR analysis

Total RNA was obtained using the Total RNA Extractor (Trizol) extraction kit. mRNA was reverse transcribed into cDNA. A random primer *p* (dN) 6, 1 μ L of dNTP Mix, and 1500 ng of RNA template were added to an RNase-free PCR tube in an ice bath. This was followed by 14.5 μ L of RNase-free ddH₂O. After the reaction mixture was mixed, it was centrifuged for 3–5 s, incubated at 65 °C for 5 min, and then incubated on ice and centrifuged. To do reverse transcription PCR, 4 μ L of 5 $\times$ RT Buffer, 0.5 μ L of RNase inhibitor, and 1 μ L of maxima reverse transcriptase were added

to an ice bath, mixed gently, and spun in a centrifuge. Ten times less cDNA was used as a template, and two times as much SG Fast qPCR Master Mix was added. After mixing evenly, PCR cycles were performed. We placed the reactants in a 96-well plate and used a fluorescence quantitative PCR machine to measure this reaction. The gene expression values were calculated using the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen 2001). Three biological replicates were performed for each sample. The primers are shown in Table S1.

Subcellular localization analysis

First, the recombinant vector was constructed by homologous recombination, which contained linker (*rDNA*11, 117 bp) and *eGFP* and *amp* resistance genes. The vector backbone was pAN580 (2021). The CDS sequence of the *WOX9* and *WOX15* genes were used and compared with the plasmid map, where the vector sequence needs to be inserted. The single restriction enzyme site was screened using Primer 5 software, and *WOX9* and *WOX15* primers were designed. Protective bases and regulatory bases were added to the primer sequence. The target gene was amplified using the BioRun Magic PCR Mix Kit, which included 1 μ L template, 2 μ L sense primer (100 μ M), 2 μ L antisense primer (100 μ M), 25 μ L Biorun Pfu PCR Mix and 20 μ L Nuclease-free water. The PCR product was subjected to agarose gel electrophoresis and gel recovery. According to the restriction enzyme site, the target gene and pAN580 vector were double-digested with BsmB1 and Esp31, mixed and centrifuged, and then kept in a water bath at 37 °C for 1 h. In order to connect the constructed vector with the target gene, 5 μ L rDNA11, 5 μ L pAN580-linker-esp31 and 10 μ L Biorun 2*EasyClone Mix were mixed evenly and connected at 37°C for 30 h. The 5–10 μ L product was added to the *E. coli* competent cells, and place it on ice for 30 min. First, it was bathed in 42 °C water for 90 s, then bathe in ice for 5 min 400 μ L of LB liquid culture medium was added and activated on a shaker for 40–50 min 200 μ L of supernatant centrifuged and discarded before blowing and mixing the remaining bacteria. 50 μ L solution was taken to coat the ampicillin-resistant plate, and inverted it in a 37 °C constant temperature incubator for overnight culture. Plaques and bacterial liquids were identified by PCR. The chosen *Nicotiana benthamiana* leaves were grown in a lab for 15–20 days. It was then soaked in five to ten milliliters of an enzyme solution. Incubate at 24 °C for enzymatic hydrolysis for 4 h. To remove the supernatant, filter it using a 40- μ m filter and centrifuge it. Twenty microliters of DNA (10 μ L of target gene vector plasmid and 10 μ L of marker plasmid) were mixed with two hundred microliters of the protoplast suspension. We then added the same amount of the PEG 4000 solution. Mix thoroughly and let it stand at room temperature for 10–15 min. To terminate the reaction, dilute it with 1 mL of W5. The protoplasts were centrifuged to collect, and the supernatant was discarded.

Table 1 GO and KEGG pathway enrichment analysis of the top 20 significantly differentially expressed genes

GO ID	Term	Ontology	Significant	Annotated	P value	Q value	KEGG ID	Description	Significant	Annotated	P value	Q value
GO:0090304	nucleic acid metabolic process	biological process	1672/5011	4649/16599	7.80E-24	6.29E-20	ko03020	RNA polymerase	38/2563	85/9638	2.29E-04	1.65E-02
GO:0051276	chromosome organization	biological process	397/5011	908/16599	5.50E-19	2.22E-15	ko03013	Nucleocytoplasmic transport	65/2563	167/9638	3.05E-04	1.75E-02
GO:0006139	nucleobase-containing compound metabolic process	biological process	1762/5011	5057/16599	5.30E-18	1.42E-14	ko04110	Cell cycle	92/2563	259/9638	8.38E-04	3.60E-02
GO:0005694	chromosome	cellular component	276/5020	603/16724	7.60E-17	9.20E-14	ko04111	Cell cycle - yeast	67/2563	179/9638	8.76E-04	3.60E-02
GO:0046483	heterocycle metabolic process	biological process	1860/5011	5406/16599	1.70E-16	2.74E-13	ko04914	Progesterone-mediated oocyte maturation	34/2563	79/9638	1.10E-03	3.95E-02
GO:0006259	DNA metabolic process	biological process	371/5011	862/16599	1.70E-16	2.74E-13	ko04212	Longevity regulating pathway - worm	53/2563	137/9638	1.24E-03	3.95E-02
GO:0006725	cellular aromatic compound metabolic process	biological process	1905/5011	5578/16599	2.00E-15	2.69E-12	ko00310	Lysine degradation	24/2563	54/9638	3.43E-03	9.87E-02
GO:0044427	chromosomal part	cellular component	229/5020	497/16724	1.50E-14	5.75E-12	ko00908	Zeaxanthin biosynthesis	16/2563	32/9638	3.90E-03	1.01E-01
GO:0043228	non-membrane-bounded organelle	cellular component	834/5020	2258/16724	1.90E-14	5.75E-12	ko04550	Signaling pathways regulating pluripotency of stem cells	8/2563	12/9638	4.22E-03	1.01E-01
GO:0043232	intracellular non-membrane-bounded organelle	cellular component	834/5020	2258/16724	1.90E-14	5.75E-12	ko00983	Drug metabolism - other enzymes	36/2563	93/9638	6.91E-03	1.33E-01
GO:0016070	RNA metabolic process	biological process	1442/5011	4116/16599	5.80E-15	6.68E-12	ko00920	Sulfur metabolism	19/2563	42/9638	6.98E-03	1.33E-01
GO:0006260	DNA replication	biological process	128/5011	238/16599	2.10E-14	2.12E-11	ko04919	Thyroid hormone signaling pathway	33/2563	84/9638	7.33E-03	1.33E-01
GO:1901360	organic cyclic compound metabolic process	biological process	1941/5011	5727/16599	3.40E-14	3.05E-11	ko00980	Metabolism of xenobiotics by cytochrome P450	22/2563	51/9638	7.64E-03	1.33E-01
GO:0006261	DNA-dependent DNA replication	biological process	105/5011	187/16599	1.10E-13	8.87E-11	ko00196	Photosynthesis - antenna proteins	12/2563	23/9638	7.87E-03	1.33E-01
GO:0034641	cellular nitrogen compound metabolic process	biological process	1958/5011	5801/16599	1.70E-13	1.25E-10	ko03018	RNA degradation	70/2563	205/9638	9.49E-03	1.52E-01
GO:0044436	thylakoid part	cellular component	180/5020	387/16724	3.70E-12	8.95E-10	ko04725	Cholinergic synapse	7/2563	11/9638	1.08E-02	1.64E-01
GO:0034357	photosynthetic membrane	cellular component	164/5020	348/16724	9.10E-12	1.84E-09	ko04320	Dorso-ventral axis formation	6/2563	9/9638	1.37E-02	1.97E-01
GO:0042651	thylakoid membrane	cellular component	163/5020	347/16724	1.50E-11	2.59E-09	ko04075	Plant hormone signal transduction	134/2563	428/9638	1.48E-02	2.03E-01
GO:0009535	chloroplast thylakoid membrane	cellular component	143/5020	300/16724	7.10E-11	9.79E-09	ko04012	ErbB signaling pathway	9/2563	17/9638	1.86E-02	2.43E-01
GO:0044428	nuclear part	cellular component	781/5020	2169/16724	7.30E-11	9.79E-09	ko04142	Lysosome	44/2563	125/9638	2.03E-02	2.54E-01

We add 1 mL of W5 and wash 1–2 times. Finally, add 1 mL of W5 solution and culture in the dark at 25 °C for 18–24 h. After taking out the supernatant, keep only 100 µL of the protoplasts and take pictures with a laser confocal microscope or fluorescence microscope.

Results

Acquisition and culture of plant materials

Based on the developmental time of calli, the callus formation was divided into 6 stages (Fig. 1). The T1 stage is the stem segment explant that has been cultured in the dark for 7 days, at which time no callus is induced. The T2 stage is the explant that has been cultured in the dark for 7 days and in the light for 8 days, at which time the explants swell and induce calluses at both ends. The T3 stage is the callus that has been cultured for 20 days, at which time all the wounds of the explants induce explants and the callus at both ends begins to grow. The T4 stage is when the callus has been cultured for 30 days. The T5 stage is when the callus has been cultured for 50 days, at which point it begins to transform into somatic embryos. The T6 stage is the tissue that has differentiated into somatic embryos (globular embryos). There were different stages of calluses that were frozen in liquid nitrogen and kept at –80 °C. At each stage, 0.5 g of callus was taken, and three biological replicates were carried out.

Identification and physicochemical properties of WOX members in *P. praeruptorum*

We analyzed the physicochemical properties of WOX family members in *P. praeruptorum* (Table 2). The amino acid lengths of WOX family members in *P. praeruptorum* range from 160 to 448 amino acids, and the molecular masses range from 18.45 to 49.92 kDa. PpWOX2, PpWOX4, PpWOX7, PpWOX10, and PpWOX12 typically contain more amino acids. The isoelectric points of all WOX family members range from 5.44 to 10.32. All members of the WOX family were predicted to be in the nucleus based on their subcellular localization. The Pfam file was utilized to screen all proteins in *P. praeruptorum*, and the CDD database from NCBI was used to verify the domain structures. Ultimately, a total of 16 family members were identified. These WOX genes were renamed based on their order on the chromosome. The amino acid sequences of the 16 PpWOX genes were aligned in multiple sequence alignments to investigate the sequence characteristics of the homologous domain of WOX. The protein sequences of the WOX gene family members contain two conserved domains: the homeodomain (HD) and the WUS-box (WB). The WOX gene family of *P. praeruptorum* exhibits a highly stable domain 1 (HD), whereas domain 2 (WB) is less stable (Figure S1).

Phylogenetic analysis of the WOX gene family

To investigate the phylogeny of the WOX family, the NJ method was used to construct a phylogenetic tree of the WOX family members: Arabidopsis (18), rice (17), tobacco

Table 2 The physicochemical properties of WOX family members in *P. praeruptorum*

Gene ID	Gene name	Sequence length	M.W	IP	Subcellular localization	Chr. position	SI	IC	Length
BHQH00039161	PpWOX1	191	22,200.16	9.2	Nucleus	Chr01	62.83	–0.871	663
BHQH00039427	PpWOX2	431	47,645.06	9.05	Nucleus	Chr02	68.75	–0.509	2495
BHQH00038834	PpWOX3	294	33,466.61	6.04	Nucleus	Chr03	50.82	–0.922	1111
BHQH00017159	PpWOX4	250	27,597.32	5.44	Nucleus	Chr03	64.64	–0.525	1968
BHQH00016987	PpWOX5	283	32,421.55	8.59	Nucleus	Chr04	67.24	–0.647	1253
BHQH00013571	PpWOX6	187	21,521.04	8.38	Nucleus	Chr04	60.91	–0.852	674
BHQH00035353	PpWOX7	242	27,282.14	9.16	Nucleus	Chr05	57.6	–0.781	1276
BHQH00017554	PpWOX8	160	18,458.19	10.32	Nucleus	Chr05	71.38	–0.938	1111
BHQH00032048	PpWOX9	239	27,366.85	9.43	Nucleus	Chr06	59.58	–1.005	1831
BHQH00006550	PpWOX10	363	40,557.48	9.32	Nucleus	Chr07	69.28	–0.619	2709
BHQH00022775	PpWOX11	368	42,182.82	6.08	Nucleus	Chr07	56.79	–0.914	2506
BHQH00023505	PpWOX12	448	49,929.98	6.76	Nucleus	Chr08	71.29	–0.45	1943
BHQH00042860	PpWOX13	281	31,527.85	5.93	Nucleus	Chr09	50.64	–0.978	3491
BHQH00036260	PpWOX14	267	30,286.04	7.24	Nucleus	Chr10	45.32	–0.872	2576
BHQH00022532	PpWOX15	290	32,984.75	9.26	Nucleus	Chr10	57.48	–0.939	1741
BHQH00008613	PpWOX16	261	29,381.34	5.94	Nucleus	Chr11	60.19	–0.944	2332

(28), maize (30), and *P. praeurptorum*. The members of the *PpWOX* family were put into groups based on the conservative features of the WOX nomenclature of *A. thaliana*. The results showed that *PpWOX*13 and *PpWOX*16 were classified as ancient clade; *PpWOX*2, *PpWOX*4, *PpWOX*7, *PpWOX*10, and *PpWOX*12 were classified as intermediate clade; *PpWOX*1, *PpWOX*3, *PpWOX*5, *PpWOX*6, *PpWOX*8, *PpWOX*9, *PpWOX*11, *PpWOX*14, and *PpWOX*15 were classified as modern clade (Fig. 2). In the intermediate clade, *PpWOX*2, *PpWOX*10, and *PpWOX*12 are closely related to *AtWOX*3 and *AtWOX*8. *AtWOX*3/*PR*S1 is primarily involved in the development of lateral organs of cells, while *AtWOX*8 plays a crucial role in the normal development of the pre-embryo (Li et al. 2020a, b). Therefore, we speculate that *PpWOX*2, *PpWOX*10, and *PpWOX*12 may have a regulatory role in the embryonic development of plants. In the ancient clade, *PpWOX*13 and *PpWOX*16 exhibit a specific

relationship with *AtWOX*10 and *AtWOX*13. Specifically, *AtWOX*13 is most strongly expressed in developing flowers and young fruits, and it plays a crucial role in promoting the formation of the embryo structure during fruit development (Deveaux et al. 2008). This suggested that *PpWOX*13 and *PpWOX*16 play significant role in flowers and fruits development. In the modern clade, *PpWOX*15 and *AtWOX*4 are closely related. *PpWOX*5 and *PpWOX*11 are closely related to *AtWOX*1 and *AtWOX*9. *AtWOX*9 is involved in maintaining cell division and preventing premature differentiation in SAM (Liu et al. 2014). Therefore, it is speculated that *PpWOX*5 and *PpWOX*11 play important roles in early growth and embryonic stages.

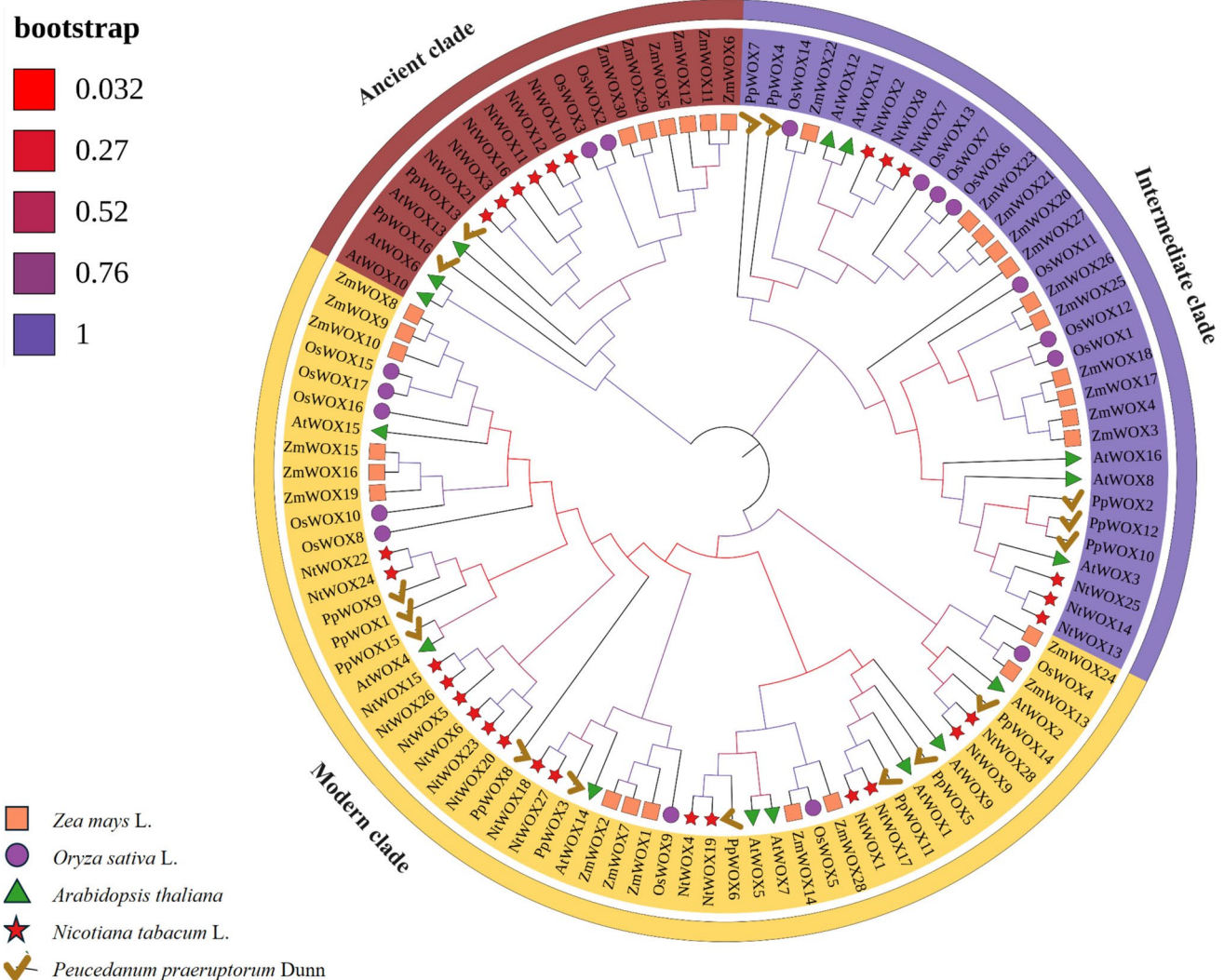


Fig. 2 Phylogenetic tree of WOX proteins among *P. praeurptorum*, *Zea mays*, *Oryza sativa*, *Arabidopsis thaliana* and *Nicotiana tabacum*

Gene structure, conserved domains and motifs of *P. praeruptorum* *WOX* members

The MEME suite was used to search for conserved motifs in 16 *PpWOX* proteins. Fifteen such motifs were found. The motif structure analysis showed that all *WOX* members in the modern clade had structures for motif 1, motif 2, motif 4, and motif 14. However, there wasn't much conservation. The ancient clades *PpWOX13* and *PpWOX16* were more conserved, and both contained the conserved motifs motif 1, motif 2, and motif 5. In the intermediate clade, *PpWOX2*, *PpWOX10*, and *PpWOX12* were more conserved and all contained motif 1, motif 2, motif 3, motif 7, and motif 8. *PpWOX4* and *PpWOX7* have high homology, and both contain motif 1, motif 2, motif 9, and motif 12 (Figure S2A). CDD search results show that *PpWOX* proteins mainly contain the homodomain superfamily and homobox-conserved domains (Figure S2B). Exon and intron analysis found that there was no obvious difference in the number of introns and exons among different family members. The number of exons ranged from 2 to 4, and only *PpWOX12* contained 5 exons. *PpWOX14* contained a longer 5'UTR sequence, while *PpWOX13* contained a longer 3' UTR sequence. *PpWOX2*, *PpWOX10*, *PpWOX11*, *PpWOX13*, and *PpWOX14* contained longer intron structures (Figure S2C).

Analysis of cis-acting elements of the *PpWOX* genes

To understand the transcriptional regulation pattern of the *PpWOX* family members, we analyzed the upstream promoters of the *PpWOX* family genes (Figure S3A). The results indicated that the promoter structure of the *Peucedanum* *WOX* gene family contained various *cis*-acting elements. The elements related to stress resistance included defense and stress response elements (4), low-temperature induction response elements (22), light response elements (206), anaerobic induction elements (28), drought induction elements (11), endosperm expression-related elements (4), and flavonoid biosynthesis-related elements (2). Auxin (4), gibberellin (9), abscisic acid (40), MeJA response elements (42), and salicylic acid response elements were the main ones that had to do with hormone regulation. The *cis*-acting elements related to material synthesis included those involved in zein metabolism regulation and endosperm expression, as well as MYB-binding elements involved in flavonoid biosynthesis genes. There are more promoter elements upstream of the *PpWOX3*, *PpWOX8*, *PpWOX9*, *PpWOX13*, *PpWOX14*, *PpWOX15*, and *PpWOX16* genes than the others. More elements are found in *PpWOX4*, which may be very important in hormone signaling pathways like abscisic acid (ABA) and methyl jasmonate (MeJA), as well as in growth and development. In addition, each *PpWOX*

contains multiple copies of LRE motif (the light response element), indicating that the *PpWOX* gene plays an important role in the light response (Figure S3B).

Analysis of colinearity and selection pressure of *PpWOX* genes

Sixteen members of the *PpWOX* family genes are spread out unevenly on 11 chromosomes, as shown by the genome data and annotations of *P. praeruptorum*. Chr03, Chr04, Chr05, Chr07, and Chr10 each have two *WOX* members, while the other chromosomes only have one member of the family. We used MCS-canX software to examine the *PpWOX* gene's colinearity with *A. sinensis* and *A. thaliana* to figure out how it has changed over time. The results indicated that there were seven whole genome duplication (WGD) or segmental duplication events in the *WOX* family members of *P. praeruptorum*, indicating that *WOX* gene expansion was obtained through large-scale gene duplication (Fig. 3A). The colinearity analysis among *P. praeruptorum*, *A. thaliana*, and *A. sinensis* suggested that there were 12 gene pairs between *P. praeruptorum* and *A. thaliana* and 24 gene pairs between *P. praeruptorum* and *A. sinensis*, which is consistent with their genetic relationship (Fig. 3B). The analysis of selection pressure showed that the Ka/Ks ratio between *PpWOX* genes was less than 1, which means that these *WOX* genes have been lost during selection. The divergence time between genes demonstrated that *PpWOX4* and *PpWOX11* diverged the earliest, about 338 MYA. *PpWOX10* and *PpWOX12* diverged the most recently, about 52 MYA (Figure S4).

Expression pattern analysis of *PpWOX* genes

Based on the expression pattern of RNA-Seq data, the expression profile of 16 *PpWOX* family members was performed during the process of callus development, formation, and differentiation. The expression patterns within the *PpWOX* genes vary widely. *PpWOX14*, *PpWOX9*, *PpWOX6*, *PpWOX15*, *PpWOX13*, and *PpWOX16* have high expression activities, among which the *PpWOX9* and *PpWOX15* genes have significantly up-regulated expression. However, *PpWOX1*, *PpWOX2*, *PpWOX3*, and *PpWOX5* are basically not expressed, and there is no obvious correlation between the expression level and the subfamily (Fig. 4).

qRT-PCR analysis of *PpWOX* genes

To further verify the transcriptome data's reliability, qRT-PCR detected the expression data of 11 *WOX* family members at six different stages of development. The results showed that *PpWOX9*, *PpWOX13*, and *PpWOX15* all had high expression levels in six different developmental stages. In particular, the expression levels of *PpWOX9* and *PpWOX15* were significantly increased during somatic

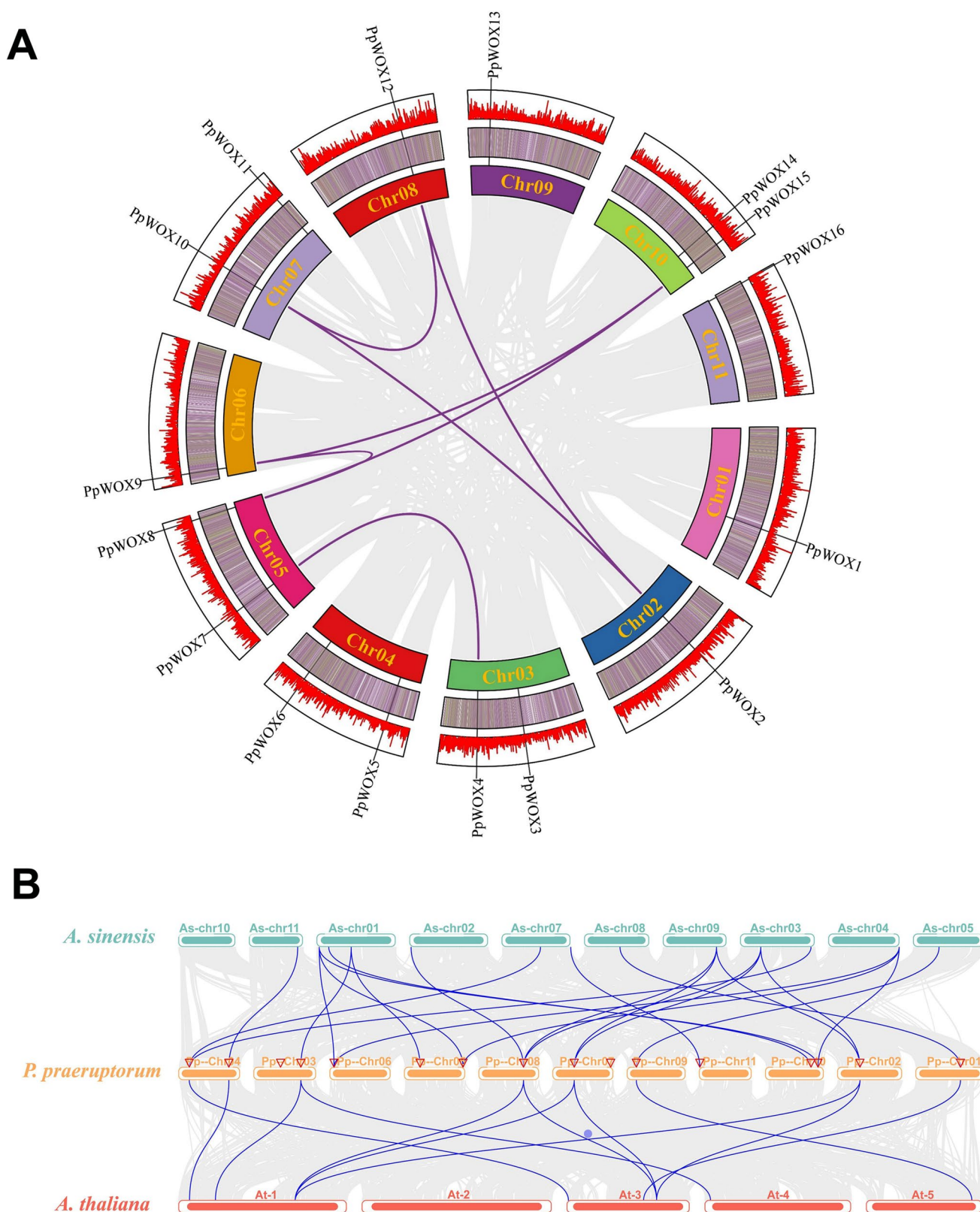
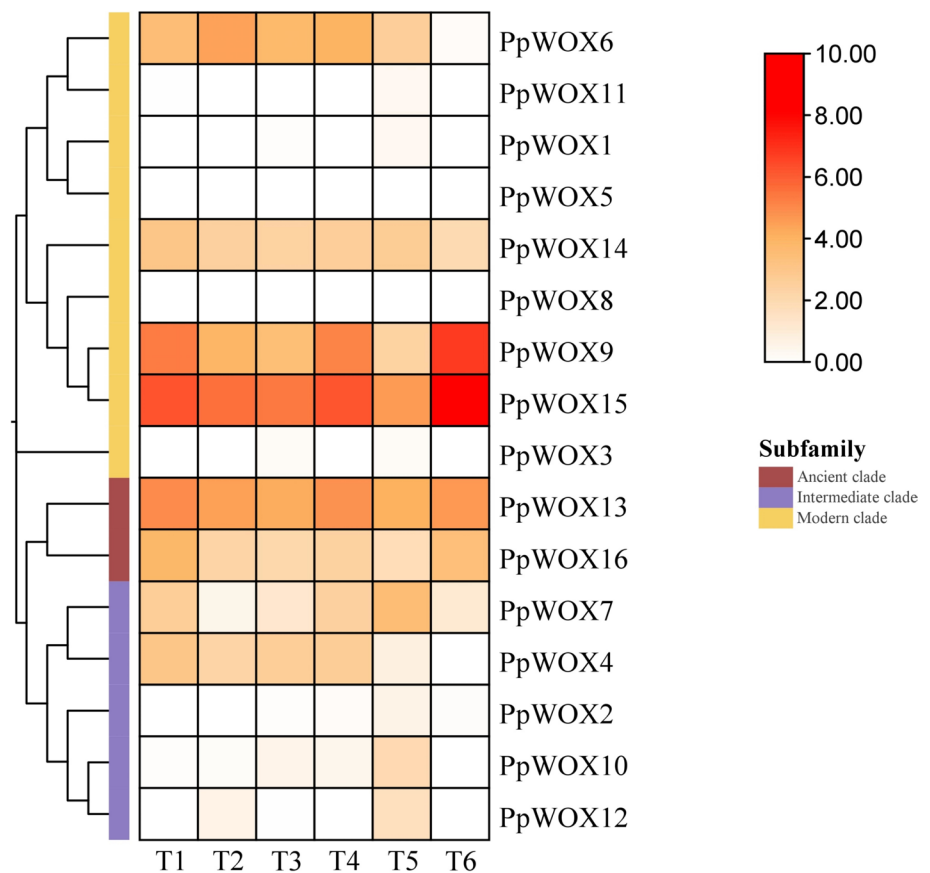


Fig. 3 Collinearity analysis and gene duplication events of the WOX genes. **A** Microsynteny analysis of *PpWOX* genes, **B** synteny analysis of WOX genes among *P. praeruptorum*, *A. sinensis*, and *A. thaliana*

Fig. 4 Expression pattern analysis of *PpWOX* genes



cell development. However, *PpWOX2*, *PpWOX4*, *PpWOX6*, and *PpWOX7* are basically not expressed in the later stages of somatic cell development. *PpWOX4* and *PpWOX7* are expressed to a certain extent in the early stages of callus development but are basically not expressed in the later stages (Fig. 5). *PpWOX2* has an opposite expression pattern to *PpWOX4*, and the expression pattern of the *PpWOX11* gene has a certain similarity with *PpWOX10*. They are specifically expressed during a critical period of callus development. This result aligns with the findings from the transcriptome data. *PpWOX9* and *PpWOX15* genes may play a regulatory role in the formation, development, and differentiation of calluses.

GO classification and KEGG enrichment analysis of differential expressed genes

The differentially expressed genes in the T2 and T6 stages of callus formation were identified based on DESeq analysis results. Significant changes in gene expression occurred during both periods. A total of 11,495 differential genes were screened in the T2 and T6 periods (Fig. 6A). The number of upregulated genes was 6,803, and the number of down-regulated genes was 4,692. There were obviously more

significantly up-regulated genes than significantly down-regulated genes. These genes affect cell division, differentiation, and development by regulating the expression of specific target genes. In the T6 stage of callus formation, *PpWOX9*, *PpWOX15*, and *PpWOX16* were significantly up-regulated, indicating that these three WOX transcription factors may be important regulators of callus differentiation. GO annotation and functional classification analysis were performed on the differentially expressed genes in the T2 and T6 stages (Fig. 5B). The differentially expressed genes included 25, 19, and 17 functional subcategories in the three major categories of biological processes, cellular components, and molecular functions, respectively. Among biological processes, the three most annotated processes are cellular processes, metabolic processes, and stimulus responses. Molecular functions account for the top three most annotated processes. KEGG enrichment on differential expressed genes was performed in the T2 and T6 stages (Fig. 6C). The results indicated that the top pathways with the most obvious enrichment are stem cell pluripotency pathways, photosynthesis-antenna protein, photosynthesis, zeatin biosynthesis, etc. After callus formation, the stem cells further differentiate according to the specific needs of the plant tissue, promoting the repair and regeneration of damaged tissue. Differentially expressed genes

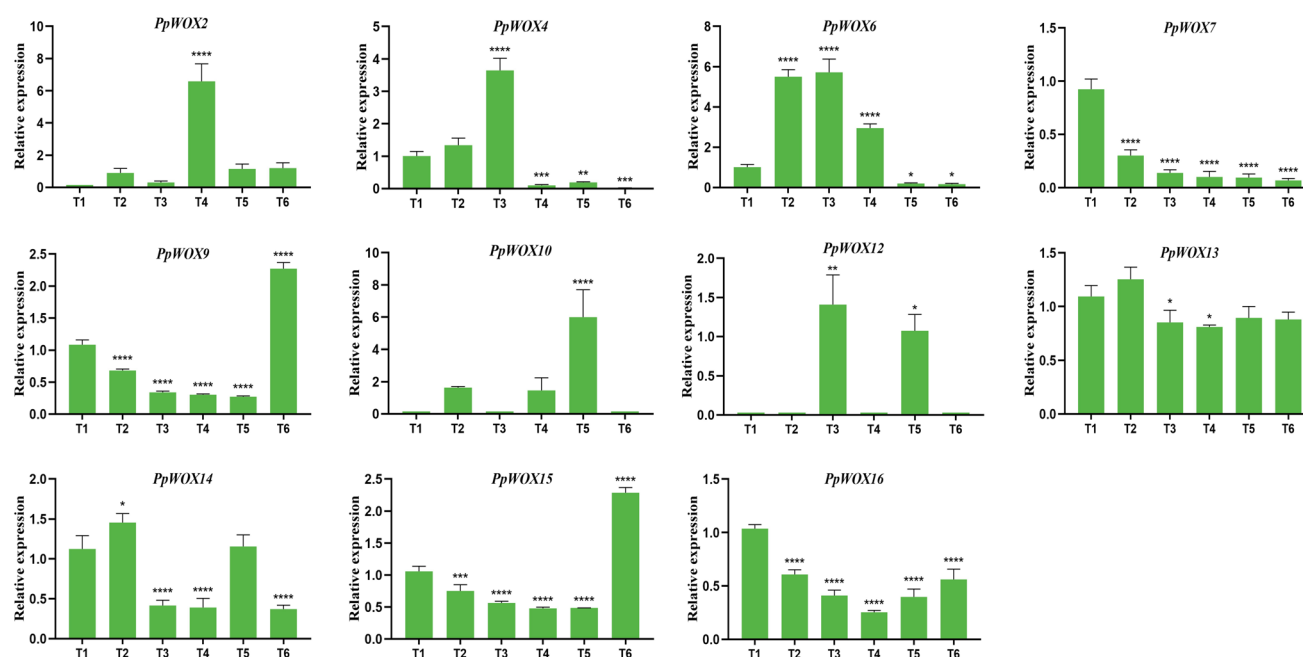


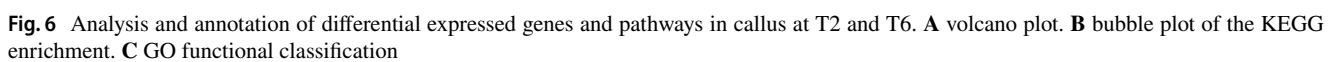
Fig. 5 qRT-PCR analysis of 11 *PpWOX* genes. T1 to T2 represent different periods of tissue development. “*”, “**”, “***” and “****” represent $P < 0.05$, $P < 0.01$, $P < 0.005$ and $P < 0.001$, respectively

were significantly enriched in the carbon fixation pathway and photosynthesis pathway of photosynthetic organisms, indicating that there are differences in photosynthesis rates at different development stages of callus, which affects the growth rate of callus.

Weighted gene co-expression network analysis of *PpWOX* genes

To construct the gene co-expression network, the RNA-seq datasets of callus tissues were chosen at different stages, and the TPM values were used as input files for WGCNA. Interestingly, 3 of the 16 *PpWOX* genes (*PpWOX9*, *PpWOX14*, and *PpWOX15*) were found to be distributed in the same module and had a certain co-expression relationship (Fig. 7A). To further study the cellular and molecular functions of the above three *PpWOX* genes, the co-expression network information of these three genes was selected, and the genes with the highest correlation with these three *PpWOX* genes were analyzed based on the inter-gene weights to construct a network diagram and visualize it. The co-expressed genes can be divided into seven clusters as a whole. Cluster 1 includes 16 genes that interact with *PpWOX9*, *PpWOX14*, and *PpWOX15*; Cluster 2 includes 22 genes that interact with *PpWOX9* and *PpWOX14*;

Cluster 3 includes 7 genes that interact with *PpWOX9* and *PpWOX15*; and Cluster 4 includes one gene that interacts with *PpWOX14* and *PpWOX15*. Cluster 5 consists of 8 genes that interact with *PpWOX9*, but not with *PpWOX14* and *PpWOX15*; Cluster 6 consists of 9 genes that interact with *PpWOX14*, but not with *PpWOX9* and *PpWOX15*; and Cluster 7 consists of 108 genes that interact with *PpWOX15*, but not with *PpWOX9* and *PpWOX14* (Fig. 7B). In addition, gene ontology (GO) and KEGG annotation were performed on all genes within the co-expression network. In GO annotation, co-expressed genes are mainly enriched in the extracellular region at the cellular component level. At the molecular functional level, they are involved in calmodulin binding, DNA-binding transcription factor activity, enzyme regulatory activity, phosphoester hydrolase activity, UDP-glycosyltransferase activity. The most enriched results are found at the biological process level, which is closely related to nutrient metabolism processes such as cell response to nutrient levels, phosphate ion homeostasis, glucose and lipid metabolism, as well as lipid and sugar metabolism (Fig. 7B). Moreover, in the KEGG pathway annotation, we identified that glycerophospholipid metabolism, lipid metabolism, plant hormone signal transduction, vitamin B6 metabolism, and other nutrient metabolism pathways were significantly enriched (Fig. 7C).



To determine the distribution of key transcription factors from the *PpWOX* family in subcellular structures and further reveal their functions, the fusion protein was constructed by recombining the coding regions of the *PpWOX9* and *PpWOX15* genes with the green fluorescent protein (GFP) gene, co-localized with red fluorescent protein (mKATE) as a nuclear localization marker, and

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the green fluorescence of GFP also proved that *PpWOX9* and *PpWOX15* were localized in the nucleus (Fig. 8).

Discussion

Plant stem cells, in addition to appearing in the shoot apical meristem (SAM) and root apical meristem (RAM) layers, also distribute throughout the body along the vasculature, such as pericytes and stems in the roots or hypocotyl of cambial cells (Aichinger et al. 2012). *WOX* genes, particularly *WUS*, play a key role in developing and maintaining stem cell niches in plants. *WUS* is expressed in the organizing center of the SAM and moves into the overlying stem cells, where it represses differentiation genes and promotes stem cell identity (Khan et al. 2024). Studying *WOX* genes reveal how plants maintain a lifelong supply of undifferentiated cells, enabling continuous organogenesis. *WOX* genes are among the first lineage-specific markers in the developing embryo. *WOX2* and *WOX8/9* are activated after zygote

division to establish the apical-basal axis, regulate asymmetric cell division, and determine regional identity (Breuninger et al. 2008; Nardmann et al. 2007). This would help address how a single cell gives rise to a complex, patterned multicellular organism. Beyond maintenance, *WOX* genes are critical for de novo organ formation. *WUS* is essential for floral meristem formation; famously, its ectopic expression can induce stem cells and entire new organs (Wang et al. 2023; Wójcik et al. 2025). Some *WOX* genes are also master regulators of adventitious root formation during cutting propagation and in response to stress (Chen et al. 2024; Safi 2024). This feature is directly applicable to plant tissue culture, micropropagation, and genetic transformation. Recent studies have shown that *WOX* genes are involved in stress resilience. *WOX11* mediates crown root development in response to waterlogging (Zhao et al. 2015a, b; Tan et al. 2025). Additionally, certain *WOX* members are induced by drought or pathogen attack, often linking stress-induced regeneration and developmental plasticity (Khan et al. 2024; Li et al. 2024). Engineering crops with enhanced resilience

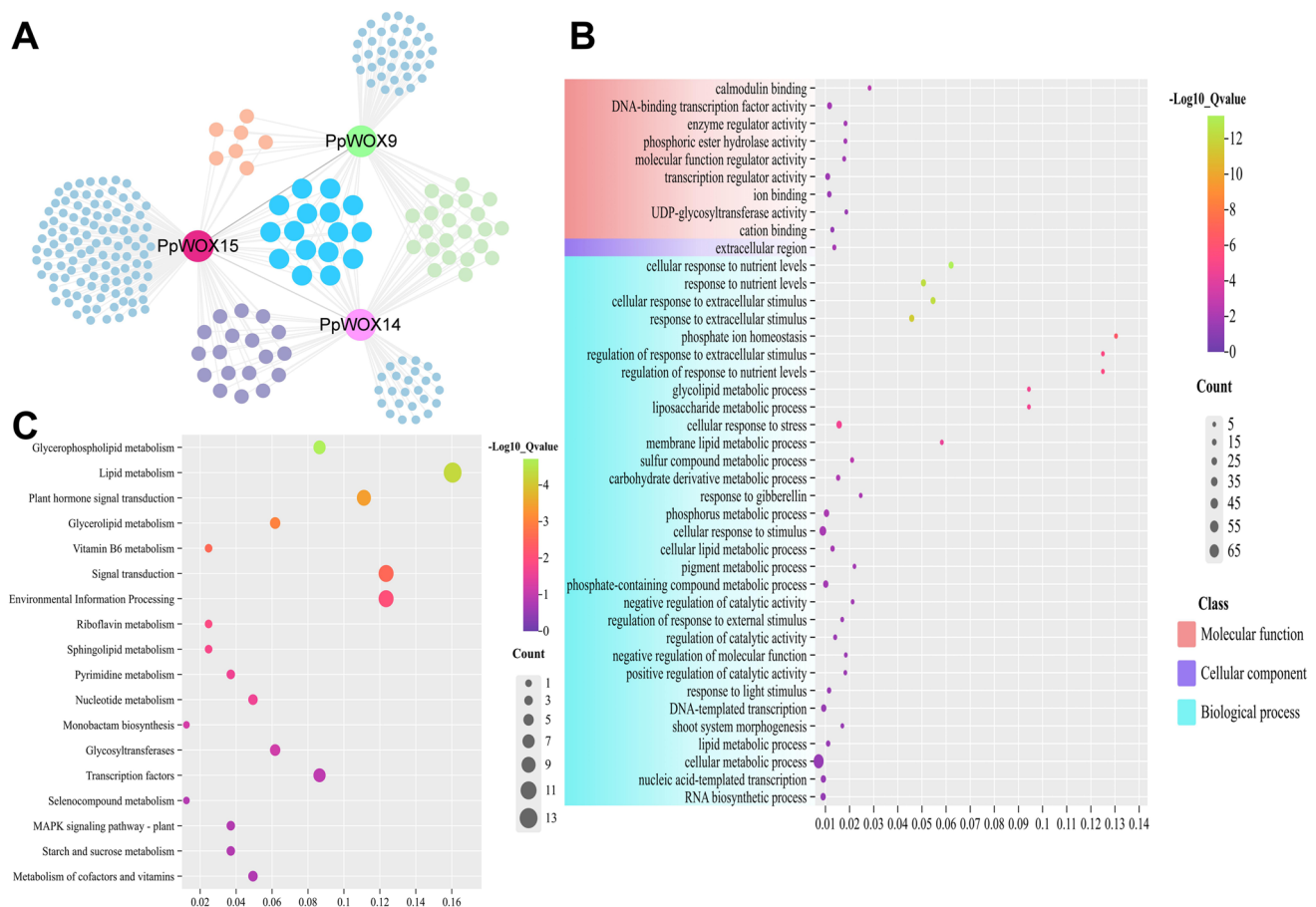


Fig. 7 The interaction network of the *PpWOX* genes based on WGCNA. **A** gene interaction analysis. Different colored dots represent genes that are co-expressed with complementary *WOX* genes, and the line connecting two nodes represents the co-expression rela-

tionship between them. The node size represents the number of *WOX* genes co-expressed with a specific gene. **B** GO enrichment analysis of co-expressed genes. **C** KEGG pathway enrichment analysis of co-expressed genes

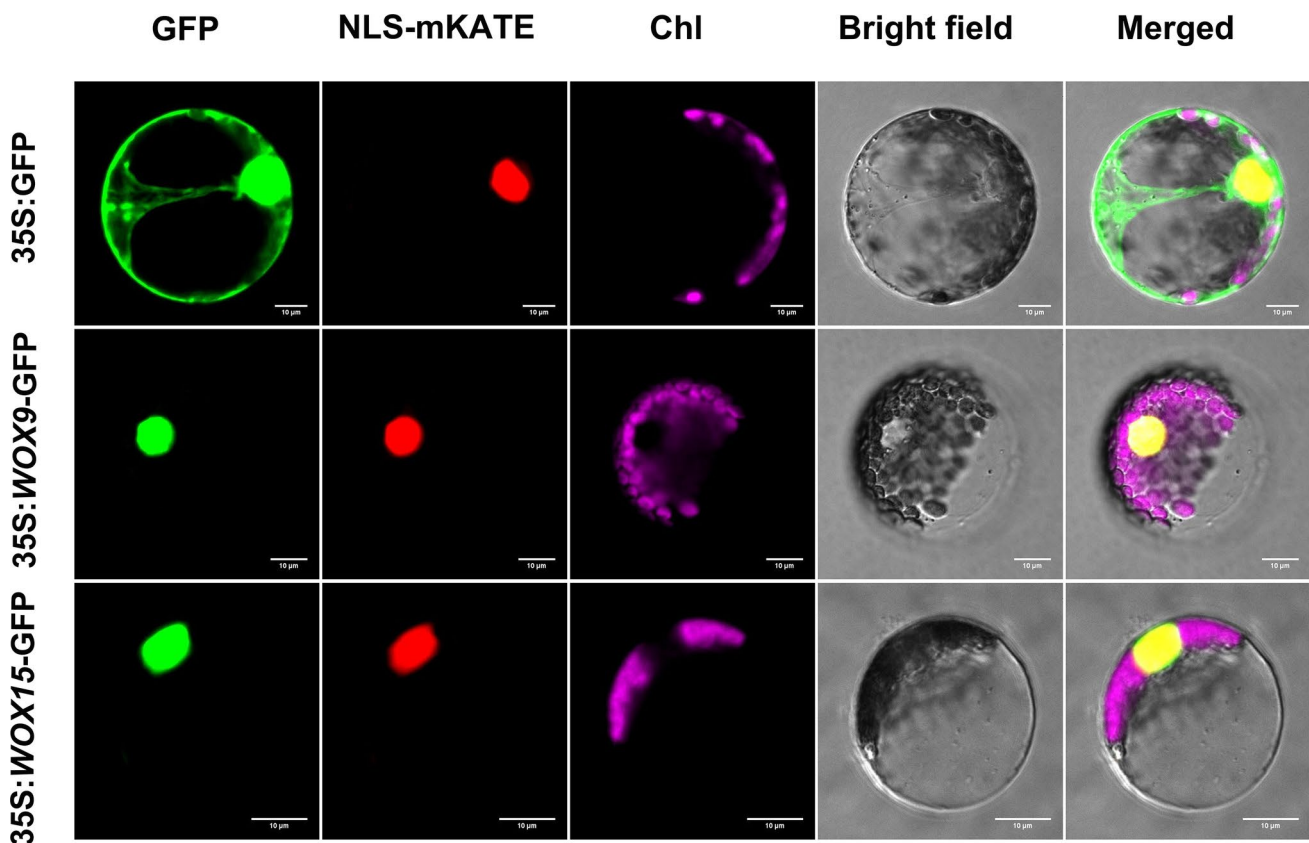


Fig. 8 Subcellular co-localization of *PpWOX9* and *PpWOX15*. 35S: GFP: empty control. GFP: green fluorescence channel. NLS:mKATE: the nuclear marker red fluorescence channel. Chl: chloroplast channel. Bright field: bright field view. Merged: overlay image

enables plants to recover more quickly from damage caused by various stresses.

Significant progress has been made regarding WOX genes' functions in model or medicinal plants (Sang et al. 2018). When studying the functions of members of the plant WOX gene family, bioinformatics methods are often used to predict and analyze gene functions, including phylogeny and colinearity analysis between different species. Through building the phylogenetic tree, it indicated that *PpWOX9* and *PpWOX15* were most closely related, where they were clustered with *AtWOX4*. *AtWOX4* has high expression activity in the root apical meristem of *A. thaliana* (Youngstrom et al. 2022). *AtWOX4* is primarily expressed in the procambium and cambium. *AtWOX4*-RNAi transgenic plants exhibit reduced stature, with an accumulation of undifferentiated cells in vascular tissue and diminished differentiation of xylem, phloem, and fiber cells (Etchells et al. 2013; Zhou et al. 2024). In poplar, both *WOX4a/b* RNA interference and double mutants exhibited markedly diminished secondary growth, accompanied by a substantially reduced vascular cambium (Kucukoglu et al. 2017; Dai et al. 2023). The xylem width of the *PtoWOX4a* overexpressing plant was also increased, despite the fact that pith development

was inhibited. Arabidopsis and woody poplar share a conserved CLE/TDIF-TDR/PXY-WOX4 regulation pathway, while WOX4 is regulated by more diverse CLEs (Li et al. 2020a, b).

Expression profiling and qRT-PCR results showed that the *PpWOX9* and *PpWOX15* genes were significantly up-regulated during callus differentiation. However, *PpWOX1*, *PpWOX9*, and *PpWOX15* also form a branch, but their expression levels are very low. This variation could be due to the loss of gene fragments during long-term gene evolution. Plant WOX proteins contain three functional domains: homeodomain (HD), WUS-box (T-L-X-L-F-P-X-X), and EAR-like motif (X-L-X-L-X-L-L). Multiple sequence alignments indicated that they contained a highly conserved HD domain. Therefore, it is speculated that the HD domain plays an important and key role in the evolution of the *PpWOX* gene family. *Cis*-acting element analysis indicated that various growth and stress response elements exist in the upstream sequences of WOX genes, including hormone response elements, light response elements, growth and development-related elements, and stress-related elements. This suggested that WOX members are not only involved in the development of somatic cells but also in responding to

external stresses and hormone signal transduction (Muhammad Tajo et al. 2022). Whole genome duplication events have played a vital role in promoting the expansion of important genes and the evolution of genetic systems (Zhu et al. 2014). Segmental duplication, tandem duplication, and retroposition are generally considered to be the three main ways of gene doubling, and tandem duplication events in plants are considered to be the main mechanism for expanding gene families (Zhang et al. 2023). In this study, seven WGD or segmental duplication events were identified in *P. praeruptorum*, suggesting that large-scale gene duplication events may play a decisive role in the expansion and functional divergency of the *WOX* genes.

Transcriptional co-expression analysis and protein interaction prediction provide the possibility for targeted screening of candidate functional genes (Urzúa-Traslaviña et al. 2021; AbuQamar et al. 2021). In this study, we attempted to identify genes or proteins that may co-express or interact with *PpWOX* genes based on big data analysis methods. The cellular and molecular effects of these co-expressed genes were further inferred by GO annotation and KEGG enrichment analysis, thereby clarifying the possible role of *PpWOX* in the development of *P. praeruptorum*. The results showed that *PpWOX9*, *PpWOX14*, and *PpWOX15* were distributed in the same module trait and exhibited a specific co-expression relationship. Most of the co-expressed genes were associated with nutrient metabolism processes, aligning with the findings from GO annotation. When the expression profile results of *PpWOX* were combined, it was revealed that *PpWOX9*, *PpWOX14*, and *PpWOX15* exhibited high expression levels and similar patterns. The GO annotation of the *PpWOX* family indicated that these genes primarily bind to DNA, thereby further regulating gene expression and cell differentiation, which promotes plant regeneration, callus formation, and organ differentiation; these regulatory functions mainly occur in the nucleus. The GO annotation results of all genes in the co-expression network of *PpWOX9*, *PpWOX14* and *PpWOX15* indicated that biological processes were enriched, closely related to the metabolism of nutrients. Therefore, *PpWOX9*, *PpWOX14* and *PpWOX15* are significantly involved in organ regeneration, callus formation, and organ development of *P. praeruptorum*, suggesting potential interactions among them. However, the existence of interactions between these transcription factors still requires verification through in vivo and in vitro experiments. *PpWOX9*, *PpWOX14* and *PpWOX15*, along with their co-expressed genes, are highly expressed in T2 and T6 periods, with the T2 period primarily related to callus formation and the T6 period related to the formation of somatic embryos.

So far, research on *WOX* regulation of plant callus formation and differentiation has primarily focused on model plants such as rice and Arabidopsis. *OsWOX11* was identified in rice callus and influences callus growth in rice

(Jiang et al. 2017). In *A. thaliana*, *AtLBD16* was activated by *AtWOX11* and specifically expressed in callus culture, suggesting that the *AtWOX11-AtLBD16* pathway is the guarantee for callus to acquire pluripotency (Liu et al. 2018). In jasmine, *WOX4*, *WOX13*, and a specific homeobox gene all play regulatory roles in callus induction and proliferation. Further studies have demonstrated that *WOX* family members exhibit multifaceted genetic interactions in callus induction (Lu et al. 2019). Regarding how *WOX* transcription factors influence plant embryogenesis, *WOX* genes have been shown to play a key regulatory role in plant embryonic development. For example, *AtWOX2* and *AtWOX8* are co-expressed in egg cells and fertilized eggs, impacting the apical and basal daughter cells of the fertilized egg, while *AtWOX9* is initiated and plays an important role in embryonic development (Wang et al. 2019). *AtWOX5* was specifically expressed in the pituitary region of most early globular embryos (Zhao et al. 2014). In this study, *PpWOX9*, *PpWOX14*, and *PpWOX15*, along with their co-expressed genes, exhibited highly expression during callus formation, particularly at T2 and T6 stages. The phylogenetic tree indicated that *PpWOX15* and *AtWOX4* are orthologous genes. *AtWOX4* is involved in the differentiation of the main stem cambium and primary roots, serving as a key regulator of main stem cambium in *A. thaliana*. *AtWOX4* mutants showed significant defects in the fascicular and interfascicular cambium of the main stem (Kucukoglu et al. 2017; Yasui et al. 2018). Therefore, *PpWOX15* may play a similar function in *P. praeruptorum* and participate in regulating the formation of the main stem cambium. It is not clear how the *WOX* transcription factor regulates and activates callus formation. There is no direct report on the mechanism of *WOX* regulation of embryonic development from an epigenetic perspective. How *WOX* regulates embryonic development still needs further exploration.

Conclusions

Based on the whole genome of *P. praeruptorum*, the *WOX* genes were identified and analyzed, including physiological property, sequence alignment, phylogenetic tree, gene structure and conserved motif, *cis*-acting element, chromosome mapping, and colinearity analyses. Callus and globular embryos of *P. praeruptorum* at different growth and development stages were collected for transcriptome sequencing analysis to determine the expression pattern of *WOX* family genes during callus formation and differentiation. The qRT-PCR analysis indicated that the expression of *WOX* genes varied in the callus at different stages. Subcellular localization results demonstrated that *PpWOX9* and *PpWOX15* were expressed in nucleus. Transcriptomic analysis was used to explore the expression pattern of *WOX* family genes during

callus formation and differentiation. It was indicated that different *WOX* genes were differentially expressed at different stages of callus formation and differentiation in *P. praeruptorum*. Specifically, *PpWOX9*, *PpWOX15*, and *PpWOX13* were highly expressed during the somatic transition. The co-expression network showed that *PpWOX9*, *PpWOX15*, and *PpWOX13* had co-expression interactions, which suggested that they might play an important role in callus formation and differentiation. Although the breeding of *P. praeruptorum* is mainly carried out through seed propagation, poor germplasm accelerates early bolting, resulting in reduced yield and quality of medicinal materials. Through in vitro tissue culture and the *Agrobacterium*-mediated genetic transformation system, these *WOX* candidates could be efficiently expressed in specific tissues and periods, which assists in cultivating high-quality seedlings. Our results provide a basis for investigating *WOX* functions in cell reprogramming and callus formation studies.

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Data availability The raw files are available in Figshare (<https://doi.org/10.6084/m9.figshare.21743984.v1>). The RNA-seq data were deposited in the SRA database with accession number PRJNA714368.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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